

EBN111 – Practical 2

NI-ELVIS : Supplementary Slides

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EBN111

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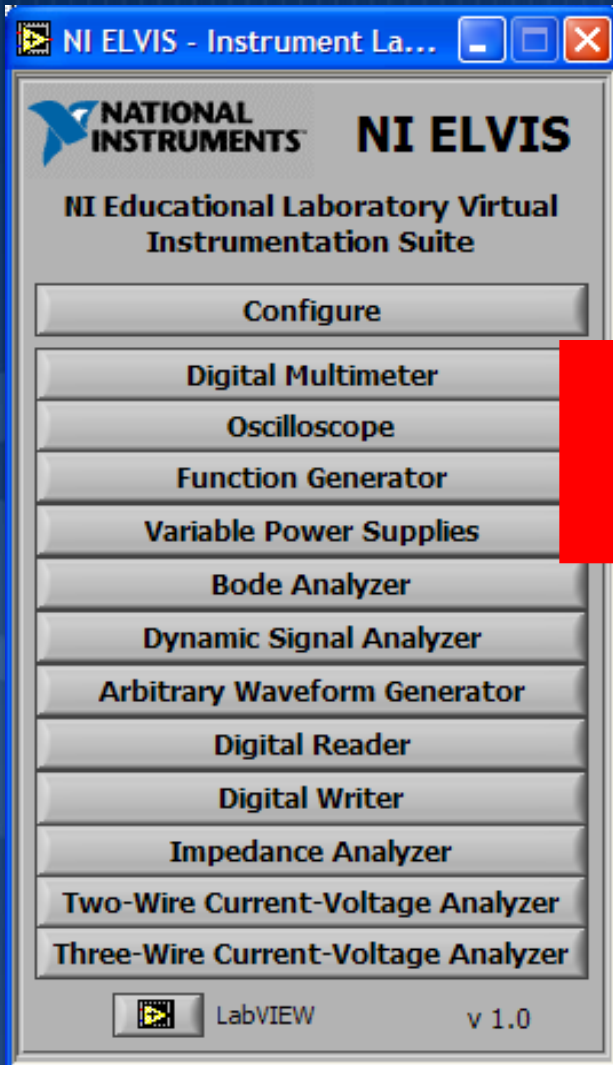
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IMPORTANT STEPS TO FOLLOW



1. Turn on system power.
2. Ensure the prototype board power is off
3. Construct the circuit
4. Connect input device
5. Connect output measurement devices
6. Turn on prototype board power
7. Launch the SFP to activate the inputs and outputs

NI-ELVIS



**Only these 4 SFP instruments
are used in the EBN122
practicum**

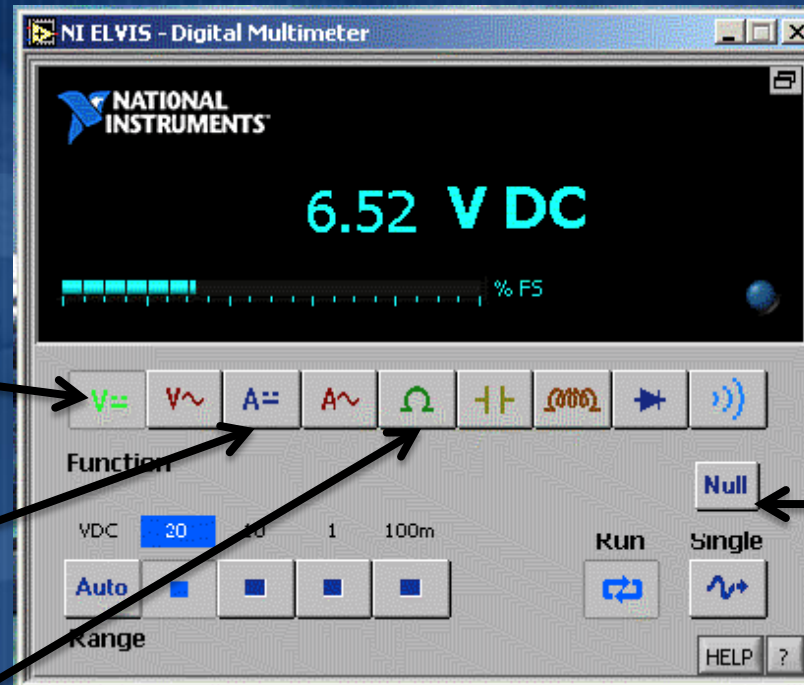
DIGITAL MULTIMETER (DMM)



Measure Voltage (DC)

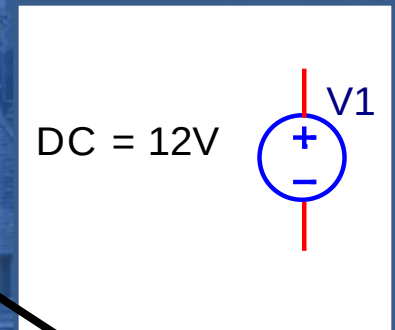
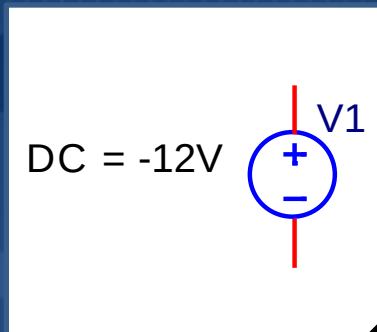
Measure Current (DC)

Measure Resistance



**NULL FEATURE
NB!!!!**

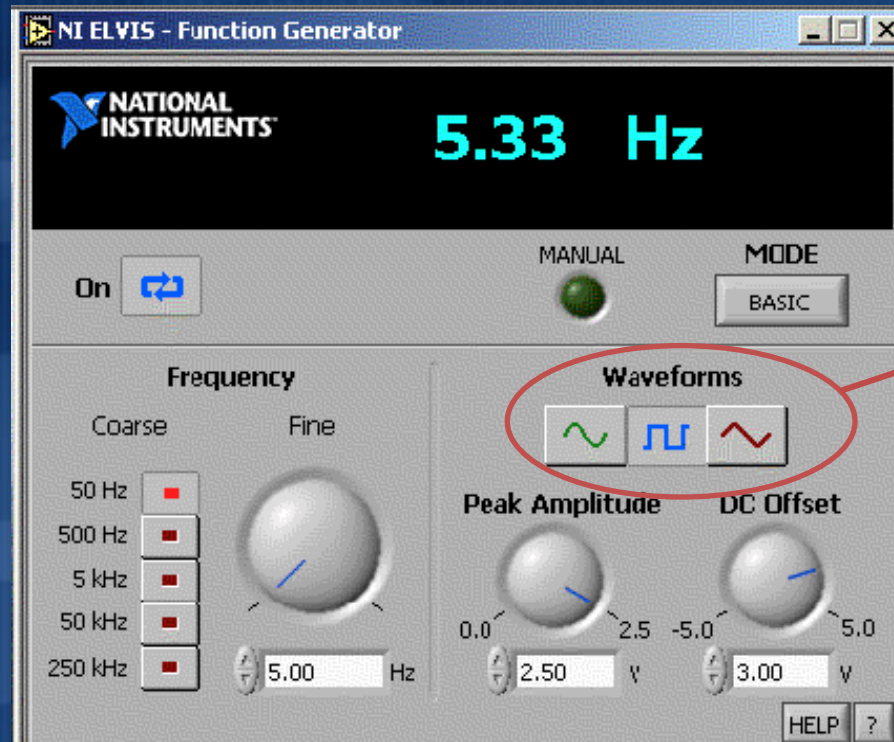
VARIABLE POWER SUPPLY



Negative Supply Voltage(DC)

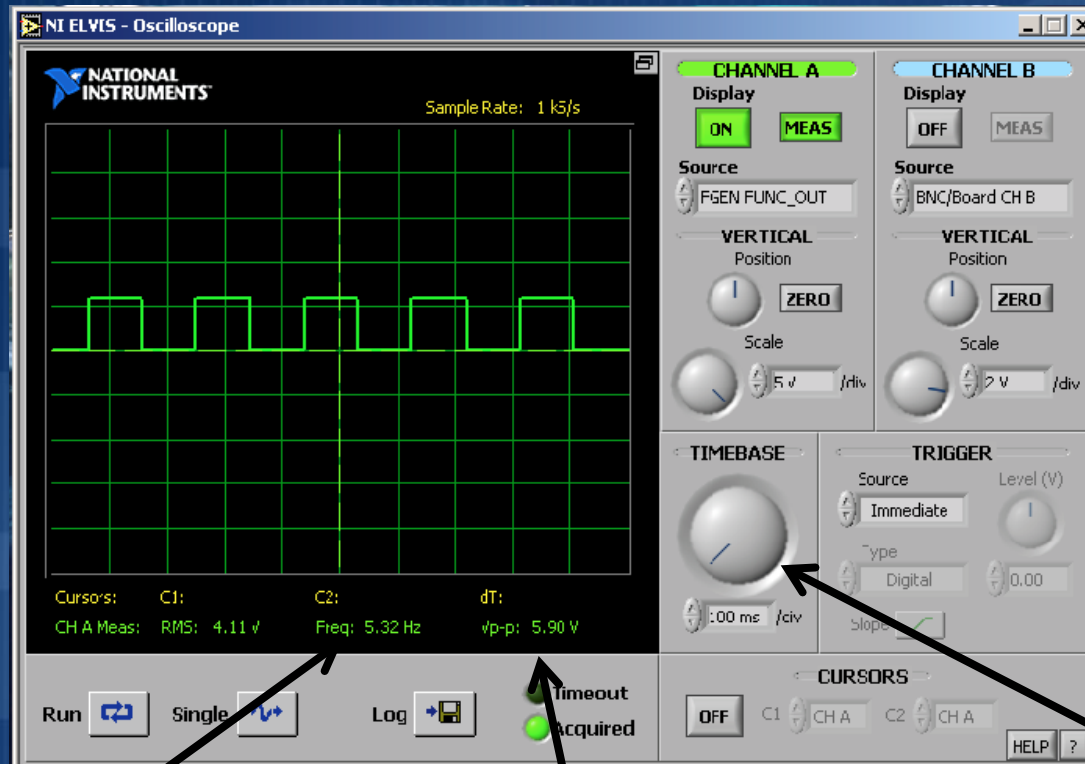
Positive Supply Voltage(DC)

FUNCTION GENERATOR



Select
Sine wave,
Square wave, or
Triangular wave

OSCILLOSCOPE

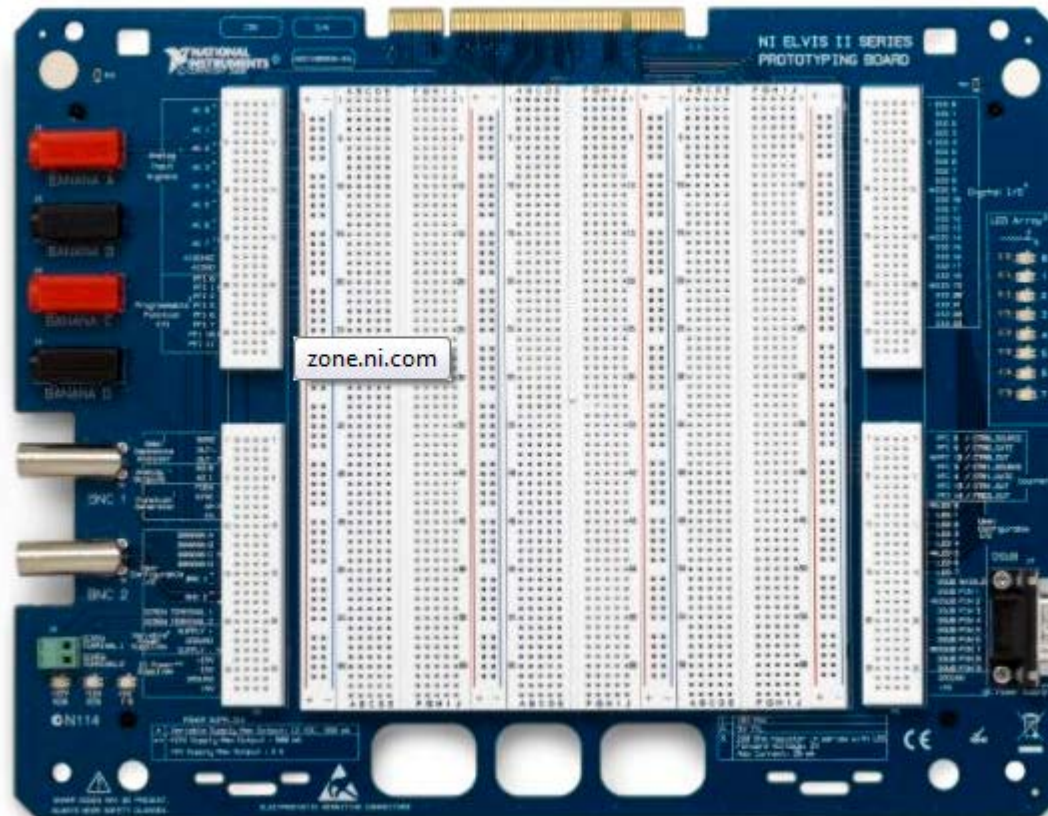


Measure frequency of signal

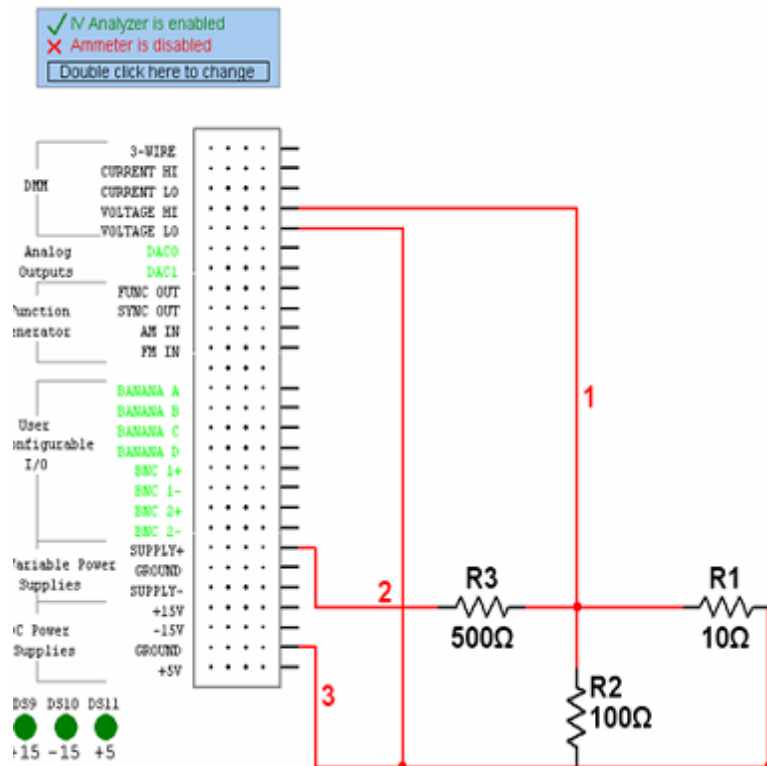
Measure peak-to-peak voltage

Scale the graph

PROTOTYPING BOARD



SAMPLE QUESTION



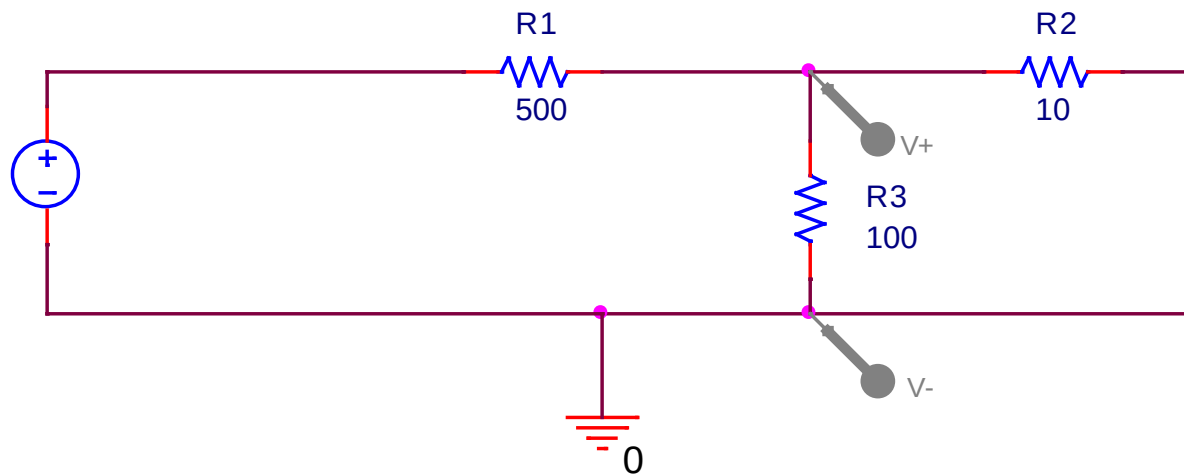
1) Voltage HI of DMM

2) SUPPLY + of variable power supply

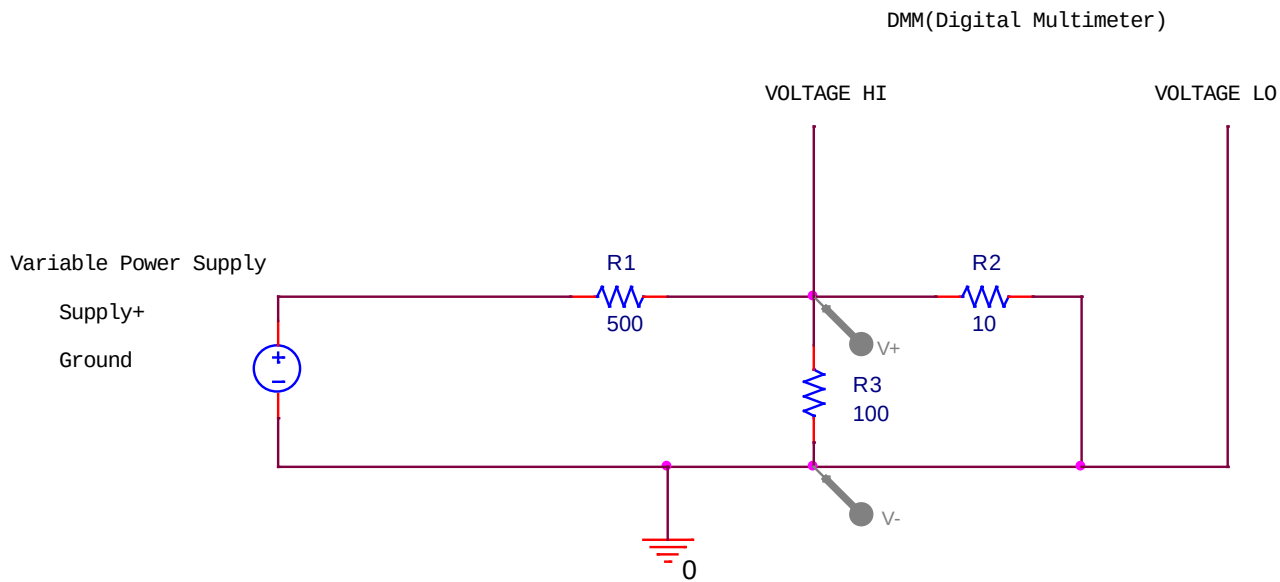
3) Ground / Voltage LO of DMM

Note VOLTAGE Hi and VOLTAGE LO are used in order to measure voltage with DMM

EQUIVALENT CIRCUIT



EQUIVALENT CIRCUIT (2)

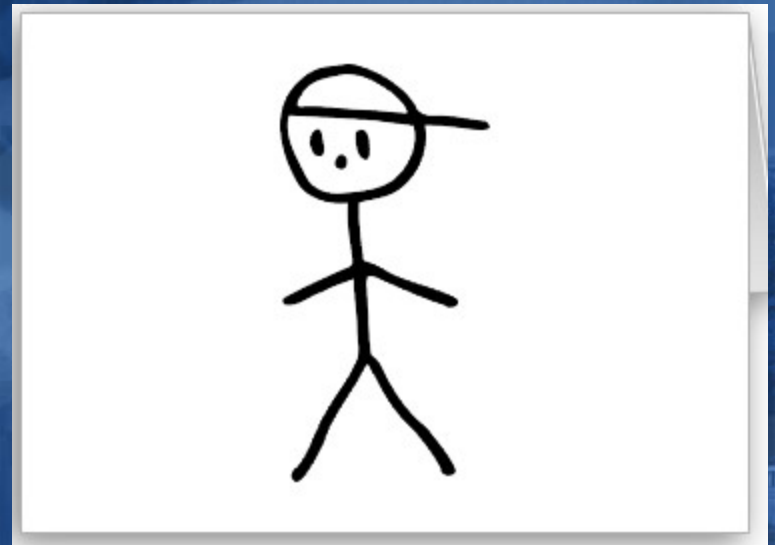
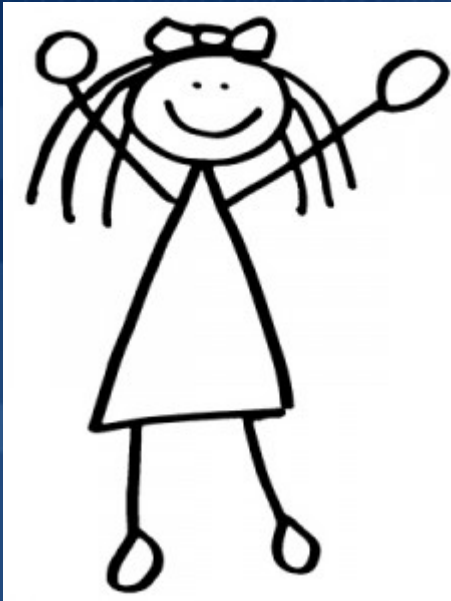


After connecting circuit correctly and starting NI-ELVIS. The DMM will display the voltage across resistor R3

NULL FUNCTIONALITY



Boyfriend wants to know how much his girlfriend weighs (he is a dorky engineer!)



Girlfriend is too shy!

NULL FUNCTIONALITY CTD...



Boyfriend tricks girlfriend by climbing on a scale with her



125 kg

He knows he weighs 70 kg... How does he calculate the girlfriend's weight???

NULL FUNCTIONALITY CTD...



HE USES THE NULL FUNCTION LIKE ON THE NI-ELVIS BOARD!!!!

Total Weight :	125 kg
Subtract Boyfriend's weight :	-70 kg

Girlfriend's weight:	55 kg

Similarly, if the Voltage on the digital multi-meter reads between 100-600 mV (due to noise) when it should be reading 0 V. The noise voltage should be NULLED!!!

Do this by pressing the NULL button on the DMM display!!

NULL EXAMPLE



Should be 0 V but is not.

NULL EXAMPLE



After clicking Null, the value is now correct

EMAIL 101



Always start an email with a greeting:

- Dear Prof Joubert,
- Dear Mr Badenhorst,
- Dear Miss Freislich,
- Geagte Prof. Joubert,
- Geagte Mr. Badenhorst
- Geagte Me. Freislich,

Always finish an email off appropriately:

Regards
Dieter Barnard

Groete
Dieter Barnard



EMAIL 101 CTD...



- Always use full sentences
- Don't use SMS language
- Take note of grammar and spelling
- Use capital letter at the start of sentences.
- Make use of paragraphs

A badly written email puts the reader in a bad mood!